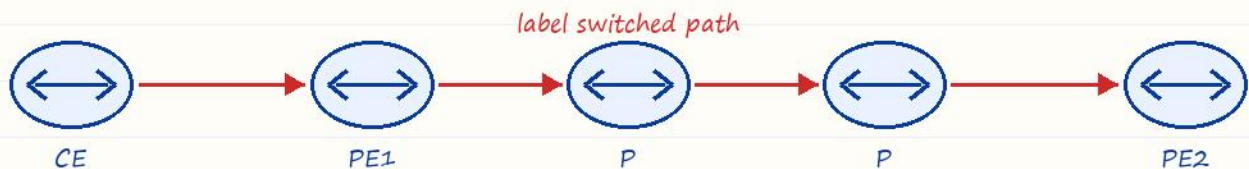


MPLS

(MULTIPROTOCOL LABEL SWITCHING)

COMPLETE NOTES



MPLS forwards packets by labels. The edge router pushes labels, core routers swap labels, and the egress pops labels to deliver the payload.

TOPICS COVERED

- * MPLS fundamentals
- * Label operations
- * LDP discovery and session states
- * Label bindings and LFIB
- * LSP verification
- * MPLS MTU, TTL, QoS
- * L3VPN labels
- * RSVP-TE and SR notes
- * Real-world troubleshooting
- * Cisco and Juniper commands

QUICK HIGHLIGHTS

- * Label = short forwarding instruction
- * FEC = forwarding equivalence class
- * LDP follows IGP reachability
- * LFIB proves programmed label actions
- * PHP implicit-null is normal
- * VPN uses outer transport + inner service label
- * MPLS adds header overhead
- * OAM validates the actual LSP
- * Control-plane up is not enough
- * Best path still starts with IGP

WHAT IS MPLS?

- Multiprotocol Label Switching forwards traffic using labels.
- It sits between Layer 2 and Layer 3 behavior.
- MPLS can transport IPv4, IPv6, L2VPN, L3VPN, and TE services.

WHY USE MPLS?

- Fast core forwarding model.
- Traffic engineering and VPN service separation.
- Clean provider-core scaling with PE/P roles.

KEY TERMS

- LER/PE: label edge router.
- LSR/P: label switch router.
- FEC: traffic class sharing forwarding treatment.
- LSP: label switched path.

CONTROL VS DATA

- Control plane distributes labels.
- Data plane pushes, swaps, and pops labels.
- Troubleshoot both planes separately.

COMMON SERVICES

- L3VPN
- L2VPN / pseudowire
- 6PE / 6VPE
- RSVP-TE
- Segment Routing over MPLS

SUMMARY

- MPLS is not a routing protocol.
- It depends on an IGP/BGP foundation.
- Labels must map to reachable FECs.

ROLES

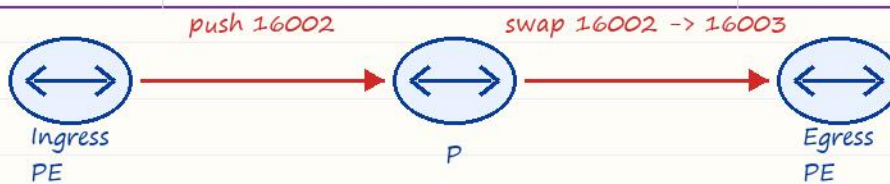
- CE peers with PE.
- PE imposes and removes labels.
- P routers swap transport labels.
- RR often carries VPNv4/VPNv6 routes.

MENTAL MODEL

- Route exists in RIB.
- Label binding exists for FEC.
- LFIB programs label action.
- OAM proves forwarding path.

KEY POINT ✨ MPLS does not replace routing. It adds a label-forwarding layer on top of reachable prefixes and service policy.

Topic	IP Routing	MPLS
Lookup	Destination IP longest match	Top label lookup
Table	RIB/FIB/CEF	LFIB / mpls.O
Core action	Forward to next hop	Swap or pop label
Ingress	No label by default	Push one or more labels
Services	VRF via route table	Inner label identifies service
Troubleshoot	show route / cef	show mpls forwarding / bindings



IP ROUTING VIEW

- Every hop checks destination IP.
- Routing table drives next hop.
- No label stack exists.

MPLS VIEW

- Ingress classifies into a FEC.
- Core follows top label.
- Egress removes label and forwards payload.

KEY POINT * The packet still has an IP payload. MPLS changes how the provider core gets it across the network.

Action	Meaning	Where	Why it matters
Push	Add label to packet	Ingress PE or service boundary	Starts LSP or adds service label
Swap	Replace top label	Transit P router	Normal core behavior
Pop	Remove top label	Penultimate or egress hop	PHP or explicit-null behavior
Lookup	Read top label	Every MPLS hop	LFIB decides next action

LABEL STACK

- Outer label carries packet across the MPLS core.
- Inner label identifies VRF, VPLS, or pseudowire at egress.
- Bottom-of-stack bit marks the last label.
- TTL, EXP/TC, and S fields ride in each label header.

SPECIAL LABELS

- 0 = IPv4 explicit-null.
- 1 = router alert.
- 2 = IPv6 explicit-null.
- 3 = implicit-null, signaled for PHP and never appears on the wire.
- 13 = GAL, common in MPLS-TP/OAM contexts.

KEY POINT ✨ When reading a trace or LFIB, translate the verb first: push starts, swap transits, pop exits.

LDP STATE MACHINE



Any fatal error tears down the session; Hello discovery starts again.

State	Meaning	Most useful check
Non-existent	No Hello	mpls ip missing, ACL, no route to peer
Initialized	TCP/open begins	TCP 646 blocked or transport address wrong
OpenSent	Init negotiation	parameter or MDS mismatch
OpenReceived	KeepAlive phase	hold-time or firewall issue
Operational	Labels exchanged	Now prove FEC bindings and LFIB

HELLO DISCOVERY

- UDP 646.
- Link Hellos discover directly connected peers.
- Targeted Hellos support non-direct sessions.
- Basic routing must exist first.

SESSION CHANNEL

- TCP 646.
- Transport address usually loopback.
- LDP router ID should be stable.
- MDS auth must match exactly.

KEY POINT * Operational LDP only proves that labels can be exchanged. It does not prove every destination has a usable LSP.

DISCOVERY CHECKS

- show mpls ldp discovery
- MPLS enabled on the interface?
- UDP 646 allowed?
- Peer Hellos visible?

TRANSPORT CHECKS

- show mpls ldp neighbor detail
- Can loopbacks reach each other?
- TCP 646 allowed?
- Transport address sane?

ROUTER ID

- Pin to loopback for stability.
- Changing it tears down sessions.
- Mismatch with IGP reachability causes pain.

AUTH

- MD5 failures may look silent.
- Check key chain/password exactly.
- Check both directions and timestamps.

FIRST LDP COMMANDS

```
show mpls ldp neighbor
show mpls ldp neighbor detail
show mpls ldp discovery
show mpls ldp parameters
telnet <peer-transport-IP> 646
```

IF DISCOVERY FAILS

- Check interface-level MPLS.
- Check ACL/multicast/security policy.
- Check directly connected subnet health.

IF TCP FAILS

- Check IGP to loopbacks.
- Check asymmetric routing.
- Check firewall/stateful inspection.

KEY POINT ✳ LDP adjacency is two problems: Hello discovery first, then TCP session establishment.

Term	Meaning	Field use
FEC	Forwarding Equivalence Class	Usually an IGP prefix for LDP
Local label	Label I advertise	Peer uses it when sending traffic to me
Remote label	Label peer advertised	I use it when forwarding to that peer
LFIB	Label forwarding table	Actual programmed label action
LIB	Label information base	Known labels, not always installed

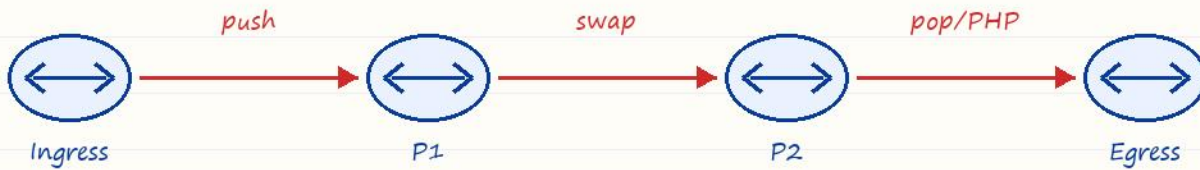
LABEL CHECKS

```
show mpls ldp bindings <prefix>
show mpls forwarding-table
show ip cef <prefix> detail
show route <prefix> detail
show mpls ldp discovery
```

READING THE RESULT

- No IGP route means no label binding for that FEC.
- A binding can exist but fail to install if next hop is not best.
- LFIB shows what hardware/software forwarding will actually do.
- Check the prefix you care about, not only neighbor state.

KEY POINT ✨ For MPLS forwarding, the real question is: does this exact FEC have the expected outgoing label and next hop?



PING MPLS

- Fast liveness proof.
- Useful before traceroute.
- Can validate a FEC/LSP quickly.

TRACEROUTE MPLS

- Hop-by-hop label path.
- Finds where label switching stops.
- Correlate with LFIB at failed hop.

TTL MODE

- Uniform mode exposes core hops.
- Pipe mode may hide the core.
- Know the mode before judging trace output.

COUNTERS

- Check interface drops.
- Check label switched packet counters.
- Large drops often point to MTU.

OAM COMMANDS

```
ping mpls ipv4 <prefix>/<mask>
traceroute mpls ipv4 <prefix>/<mask>
show mpls forwarding-table <prefix>
show interfaces counters errors
```

KEY POINT * A healthy control plane is a claim. MPLS OAM and counters prove whether labeled packets actually survive the path.



Item	Purpose	Where to check
Outer label	Gets packet across the provider core	LDP, RSVP-TE, or SR-MPLS
Inner label	Identifies VRF/service at egress PE	VPNv4/VPNv6 or L2VPN signaling
Route target	Controls VRF import/export	BGP extended community
RD	Makes overlapping prefixes unique	VPN route distinguisher

WHEN CORE MPLS IS FINE

- Check VPNv4/VPNv6 BGP session.
- Check route-target import/export.
- Check VRF route table.
- Check inner service label.

WHEN VRF TRAFFIC FAILS

- CE routing may be wrong.
- RT policy may block import.
- Egress PE may lack service label.
- MTU may fail only for large packets.

KEY POINT * Core LSP working does not mean VPN working. L3VPN needs transport reachability and the correct VPN/service label.

Concept	Meaning	Troubleshoot
RSVP session	Signals TE LSP state	Does not prove CSPF has usable path forever
CSPF	Constrained SPF	Checks constraints like bandwidth and explicit path
ERO	Explicit Route Object	Defines the path the LSP should follow
Bandwidth	Reservation constraint	May fail if unavailable on any hop
FRR	Fast reroute	Local protection for link/node failure

RSVP-TE DOWN

- CSPF found no path.
- Explicit path hop unreachable.
- Bandwidth unavailable.
- Admin group/color constraint excludes links.

RSVP-TE UP BUT WRONG

- Policy steering missing.
- IGP metric changed.
- Fast-reroute detour active.
- OAM follows unexpected path.

TE CHECKS

```
show mpls traffic-eng tunnels
show rsvp session
show mpls forwarding-table tunnels
traceroute mpls traffic-eng tunnel <id>
```

KEY POINT * RSVP state up is not the same as 'the right traffic follows the right engineered path'. Verify steering and OAM.

SR-MPLS

- Uses labels as Segment IDs.
- Headend imposes a segment list.
- Core avoids per-policy RSVP state.

LDP

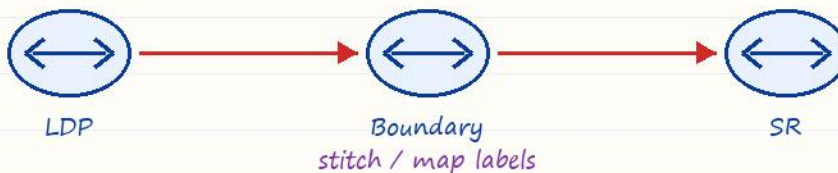
- Distributes labels for IGP FECs.
- Follows shortest path by default.
- Session is peer-to-peer.

INTERWORKING

- Maps LDP labels and SR labels at boundary.
- Useful for gradual migration.
- IGP reachability remains the foundation.

MIGRATION

- Keep SRGB consistent.
- Verify label stack depth.
- Move domains deliberately.
- Prove OAM before and after.



SRGB GOTCHA

- Inconsistent SRGB makes labels harder to reason about.
- Use consistent index planning where possible.

OAM GOTCHA

- Classic MPLS ping/traceroute support varies.
- Use SR-specific OAM where needed.

KEY POINT ✨ LDP-to-SR migration should not be a flag day. Interworking exists so you can move the core in controlled pieces.

MTU

- Each label adds 4 bytes.
- VPN/SR stacks add more overhead.
- Large packets may fail while small pings pass.
- Test with size and DF bit.

TTL

- Uniform mode propagates TTL.
- Pipe mode hides the core.
- Traceroute behavior depends on mode.

QOS

- EXP/TC bits carry class treatment.
- QoS propagation policy matters at boundaries.
- PHP can affect where markings are observed.

FRAGMENTATION

- Avoid core fragmentation.
- Set MPLS MTU end-to-end.
- Check customer-facing MTU too.

MTU / QOS CHECKS

```
show mpls interfaces
show interface <int> | include MTU
ping <dest> size <n> df-bit
show mpls forwarding-table detail
show policy-map interface <int>
```

SYMPTOM

- Small ping works.
- Application or full-size packets fail.
- Only MPLS path breaks.

LIKELY CAUSE

- Label stack exceeds path MTU.
- Core interface MPLS MTU inconsistent.
- QoS/policer drops labeled traffic.

KEY POINT ✨ MPLS MTU failures are sneaky because control-plane sessions and tiny test packets can look perfect.

MPLS TROUBLESHOOTING WORKFLOW

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Symptom	Check first	Best command
No LDP neighbor	Discovery, MPLS interface, UDP/TCP 646	show mpls ldp discovery
Neighbor stuck	Transport, TCP 646, MD5, hold-time	show mpls ldp nei detail
No label	IGP route to FEC, label filters	show mpls ldp bindings
Blackhole	LFIB, CEF, next hop, stale label	show mpls forwarding-table
Large packets fail	MPLS MTU / stack depth	ping size df-bit
VPN route missing	RT import/export, VPNv4, VRF	show bgp vpnv4

CAPTURE FIRST

- Neighbor detail.
- Bindings for exact prefix.
- LFIB entry.
- Traceroute MPLS output.

CHANGE SECOND

- Fix one likely cause.
- Wait for reconvergence.
- Recheck RIB, labels, LFIB, OAM.

KEY POINT * Troubleshoot MPLS by FEC. A global 'MPLS is up' view hides prefix-specific label and forwarding failures.

Scenario	Likely cause	Check
LDP up, traffic drops	No label for FEC, stale LFIB, MTU	bindings + forwarding table
PE loopback unreachable	IGP issue, transport address mismatch	route to peer loopback
VPN route present no traffic	Inner label, RT, CE routing	VRF route + VPN label
Traceroute dies mid-core	Label swap missing or stale	LFIB at failed hop
Only large flows fail	MPLS MTU / policer	DF size test
RSVP tunnel down	CSPF, bandwidth, explicit path	TE tunnel detail

FIELD HABITS

- Always test the exact prefix/FEC.
- Compare ingress and transit LFIB.
- Correlate changes with IGP events.
- Check MTU before blaming applications.

ESCALATION PACKAGE

- Topology and affected FEC.
- Before/after show outputs.
- MPLS traceroute.
- Recent circuit or policy changes.

MUST-HAVE OUTPUTS

```
show mpls ldp neighbor detail
show mpls ldp bindings <prefix>
show mpls forwarding-table <prefix>
traceroute mpls ipv4 <prefix>/<mask>
```

KEY POINT ✨ Most MPLS incidents are still routing, label binding, LFIB programming, or MTU. Keep the fault tree simple.

BASIC LDP

```
ip cef
mpls label protocol ldp
mpls ldp router-id Loopback0 force
!
interface Loopback0
 ip address 10.0.0.1 255.255.255.255
!
interface GigabitEthernet0/0
 ip address 10.0.12.1 255.255.255.252
 mpls ip
```

L3VPN CLUES

```
ip vrf CUST-A
 rd 65000:10
 route-target both 65000:10
!
router bgp 65000
 address-family vpnv4
  neighbor 10.0.0.2 activate
  neighbor 10.0.0.2 send-community extended
 address-family ipv4 vrf CUST-A
```

IOS / IOS-XE SHOW COMMANDS

```
show mpls ldp neighbor
show mpls ldp discovery
show mpls ldp bindings <prefix>
show mpls forwarding-table
show ip cef <prefix> detail
show bgp vpnv4 unicast all
```

CONFIG REVIEW

- CEF enabled.
- Router ID stable.
- MPLS enabled on every core-facing interface.
- IGP carries loopbacks.

CHANGE WARNING

- Router-ID change resets LDP.
- Removing mpls ip breaks LSP transit.
- Changing VRF RT changes route visibility.

KEY POINT ✨ The most common IOS miss is configuring LDP globally but forgetting mpls ip on the interface.

BASIC LDP

```
mpls ldp
router-id 10.0.0.1
interface TenGigE0/0/0/0
!
address-family ipv4
!
router ospf CORE
area 0
interface Loopback0
interface TenGigE0/0/0/0
network point-to-point
```

TE / SR CLUES

```
mpls traffic-eng
interface TenGigE0/0/0/0
!
segment-routing
global-block 16000 23999
!
router isis CORE
address-family ipv4 unicast
segment-routing mpls
```

IOS-XR SHOW COMMANDS

```
show mpls ldp neighbor
show mpls ldp bindings
show mpls forwarding
show route <prefix> detail
show cef <prefix> detail
show mpls traffic-eng tunnels
```

XR GOTCHA

- Address-family hierarchy matters.
- Interface inheritance can hide config.
- Use commit history during incident review.

VERIFY

- Route to peer loopback.
- LDP interface enabled.
- LFIB entry for exact prefix.
- OAM across the LSP.

KEY POINT ✨ On IOS-XR, remember the hierarchy: protocol, address-family, area/interface, and committed state.

BASIC MPLS / LDP

```
set protocols mpls interface ge-0/0/0.0
set protocols ldp interface ge-0/0/0.0
set protocols ldp transport-address 10.0.0.1
set protocols ospf area 0 interface lo0.0 passive
set protocols ospf area 0 interface ge-0/0/0.0
set interfaces ge-0/0/0 unit 0 family mpls
set interfaces ge-0/0/0 unit 0 family inet address 10.0.12.1/30
```

JUNOS CHECKS

```
show ldp session
show ldp neighbor
show ldp database
show route table mpls.0
show route <prefix> detail
traceroute mpls ldp <prefix>
```

JUNOS TABLES

- *inet.0*: normal IPv4 routes.
- *inet.3*: MPLS next-hop resolution in many VPN designs.
- *mpls.0*: label switching table.

JUNOS GOTCHAS

- family mpls required on MPLS interfaces.
- Policy controls route export/import.
- Check hidden routes when expected route is absent.

VPN CHECKS

- *show route table <vrf>.inet.0*
- *show route table bgp.l3vpn.0*
- *show route advertising-protocol bgp*
- *show route receive-protocol bgp*

OAM CHECKS

- *ping mpls ldp*
- *traceroute mpls ldp*
- *show rsvp session*
- *show mpls lsp extensive*

KEY POINT ✨ On Junos, check family mpls and mpls.0 early. A missing family config can make the core look haunted when it is just not MPLS-enabled.

PRODUCTION CHECKLIST

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BEFORE CHANGE

- Snapshot LDP/RSVP/SR sessions.
- Capture bindings for key FECs.
- Capture LFIB for key prefixes.
- Confirm IGP and BGP baselines.

DURING CHANGE

- Change one domain or link group at a time.
- Watch session flaps and label churn.
- Run MPLS ping/traceroute on test FECs.
- Monitor interface errors and MTU symptoms.

ROLLBACK TRIGGERS

- Unexpected mass LDP reset.
- VPN route import failure.
- LSP OAM fails for critical prefixes.
- Core packet loss or MTU failures.

AFTER CHANGE

- Compare route and label counts.
- Validate VPN/service traffic.
- Save final show outputs.
- Document any intentional path change.

BASELINE COMMAND SET

```
show mpls ldp neighbor
show mpls forwarding-table summary
show mpls ldp bindings <critical-prefix>
traceroute mpls ipv4 <critical-prefix>/<mask>
show bgp vpnv4 unicast all summary
```

KEY POINT ✨ Good MPLS changes have before/after evidence for sessions, labels, LFIB, OAM, and service routes.

QUICK COMMAND REFERENCE

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Focus	Command	Question
LDP neighbor	show mpls ldp neighbor	Is session Operational?
Discovery	show mpls ldp discovery	Are Hellos visible?
Bindings	show mpls ldp bindings <prefix>	Labels for exact FEC?
LFIB	show mpls forwarding-table	Programmed action?
CEF	show ip cef <prefix> detail	Imposition next hop?
OAM	tracertoute mpls ipv4 <prefix>/<mask>	Where does LSP fail?
Interface	show mpls interfaces	MPLS enabled and MTU?
Parameters	show mpls ldp parameters	Router ID and timers?

JUNIPER

- show ldp session
- show ldp database
- show route table mpls.0
- tracertoute mpls ldp <prefix>

IOS-XR

- show mpls ldp neighbor
- show mpls forwarding
- show route <prefix> detail
- tracertoute mpls ipv4 <prefix>/<mask>

KEY POINT * Use the command that answers the next question in the chain. Neighbor, binding, LFIB, OAM, counters.



Lab breaks: remove mpls ip, break loopback route, alter MTU, filter labels, change RT import.

LAB TASKS

- Build OSPF/IS-IS underlay.
- Enable LDP on transit interfaces.
- Verify labels per PE loopback.
- Add one L3VPN and test CE reachability.

BREAK/FIX TASKS

- Block TCP 646.
- Remove route to peer loopback.
- Create MTU mismatch.
- Filter label advertisement.

BEST PRACTICES

- Stable loopbacks for IDs.
- Consistent MTU and SRGB.
- Document RT/RD policy.
- Prove OAM per critical service.
- Keep control/data-plane checks separate.

GOLDEN RULE

- If the exact FEC has no route, it gets no useful label.
- If the LFIB has no action, the data plane will not magically work.
- If large packets fail, suspect MTU early.

KEY POINT ✨ MPLS troubleshooting gets simple when you ask: route exists, label exists, LFIB action exists, packet survives.